

SiamRAM: Siamese Tracking with Distractor-Aware Memory and Multi-Phase Reacquisition

System Description

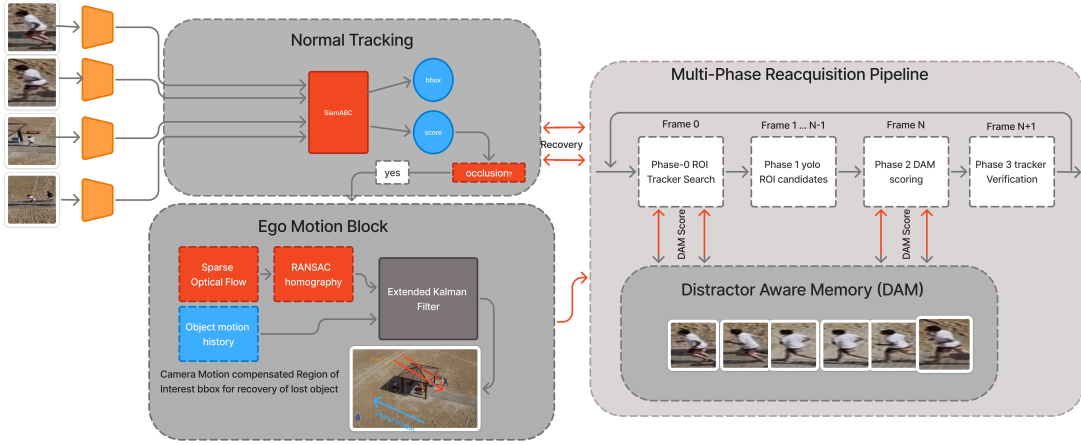


Figure 1: Overall architecture of SiamRAM. Three tightly coupled subsystems: the SiamABC core tracker (top-left), the Ego-Motion Block for camera-motion-compensated Kalman prediction (bottom-left), and the Multi-Phase Reacquisition Pipeline backed by the Distractor-Aware Memory (right).

1 Introduction

Single-object tracking fails in two recurring situations: temporary occlusion and the presence of distractors—objects with a similar appearance to the target. Lightweight trackers drift under occlusion and swap to distractors; heavy memory-augmented systems address both at the cost of real-time latency.

SiamRAM unifies three complementary mechanisms around the SiamABC [4] core tracker: (1) a *Distractor-Aware Memory (DAM)* with a short-term RAM buffer and a stable DRM bank; (2) an *Ego-Motion Block* that compensates for camera motion via sparse optical flow, RANSAC homography, and an EKF [7]; and (3) a *Multi-Phase Reacquisition Pipeline* that cascades through Siamese search, YOLO detection, and DAM-scored verification to recover the target after occlusion. Section 2 describes the core tracker, Section 3 the memory module, and

Section 4 occlusion recovery.

2 Normal Tracking with SiamABC

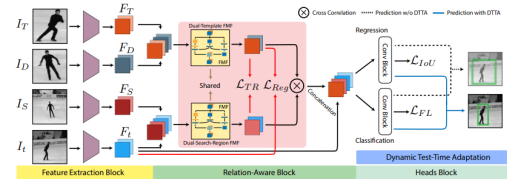


Figure 2: SiamABC normal tracking block. Four crops—static template, dynamic template, search region, and dynamic search—are encoded and fused through attention before the convolutional heads produce the classification and regression maps.

2.1 Core Tracker Architecture (Siam-ABC)

SiamABC [4] takes four inputs per forward pass, depicted in Fig. 2: a static template z (first-frame crop), a dynamic template $\tilde{z}$ updated online, a search region x centred on the current estimate, and a dynamic search $\tilde{x}$ from the most recent high-confidence memory frame. All four crops share the same ResNet backbone $\mathcal{E}$ followed by a channel-reduction neck $\mathcal{N}$:

$$f_\bullet = \mathcal{N}(\mathcal{E}(\bullet)), \quad \bullet \in \{z, \tilde{z}, x, \tilde{x}\}. \quad (1)$$

Template and dynamic-template features are concatenated and passed through a Fast Parallel Polarized Self-Attention module $\mathcal{A}$ with a second neck $\mathcal{N}'$, and analogously for search:

$$m_z = \mathcal{N}'(\mathcal{A}([f_z; f_{\tilde{z}}])), \quad m_x = \mathcal{N}'(\mathcal{A}([f_{\tilde{x}}; f_x])). \quad (2)$$

The two mixed-attention maps feed the BoxTower head $\mathcal{T}$, producing a regression map $\hat{\Delta}$ and a classification map $\hat{c}$:

$$(\hat{\Delta}, \hat{c}) = \mathcal{T}(m_x, m_z). \quad (3)$$

The final box is decoded at the highest-scoring location in $\hat{c}$, with a cosine penalty window suppressing boundary responses. To improve robustness against distractors, the BoxTower head is additionally fine-tuned on hard negatives—pairs where a distractor lies within the search region—increasing the discriminative sharpness of $\hat{c}$ without modifying the backbone.

2.2 Dynamic Template Update

SiamRAM maintains a rolling window $\mathcal{M}$ ($M=20$ frames) of high-quality frames. A frame (I_k, b_k) is admitted when:

$$(s_k \geq \hat{\rho}_k \text{ or } s_k \geq \tau_u) \text{ and } \text{IoU}(b_k, b_{k-1}) \geq \tau_{\text{iou}}, \quad (4)$$

where $\tau_u=0.87$, $\tau_{\text{iou}}=0.6$, and $\hat{\rho}_k$ is a running EMA ($\lambda=0.1$) of past classification scores. Every $N=10$ frames the best stored frame replaces the dynamic template $\tilde{z}$, provided it exceeds τ_u .

2.3 Adaptive Batch Normalization at Test Time

Select encoder normalization layers adapt their running statistics incrementally to each incoming search crop [6]:

$$\mu_\ell \leftarrow (1-\alpha)\mu_\ell + \alpha\hat{\mu}_\ell, \quad \sigma_\ell^2 \leftarrow (1-\alpha)\sigma_\ell^2 + \alpha\hat{\sigma}_\ell^2, \quad (5)$$

while learned affine parameters γ_ℓ, β_ℓ remain frozen. This provides implicit test-time adaptation to illumination and sensor-noise shifts at negligible runtime cost.

3 Distractor-Aware Memory (DAM)

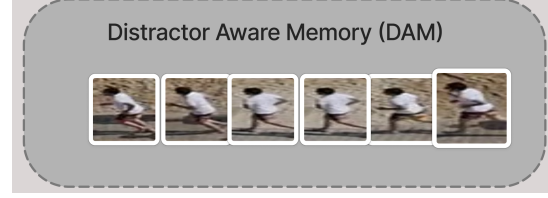


Figure 3: Distractor-Aware Memory. RAM stores geometrically verified short-term hypotheses; sufficiently stable entries are promoted to the DRM bank to anchor appearance scoring during recovery.

The DAM module, shown in Fig. 3, maintains two lightweight descriptor buffers [5]: the RAM (capacity $C_R=50$) holds a short geometric history, and the DRM (capacity $C_D=20$) stores curated stable anchors for long-occlusion recovery.

3.1 Appearance Descriptor

Given a frame I_t and box $b=(x, y, w, h)$, a patch P is extracted and two independently L2-normalised components are computed: a 16×16 grayscale texture vector $\hat{\mathbf{p}} \in \mathbb{R}^{256}$ and an $8 \times 4 \times 4$ HSV histogram $\hat{\mathbf{h}} \in \mathbb{R}^{128}$. They are combined and re-normalised:

$$\phi(I_t, b) = \frac{[w_g \hat{\mathbf{p}}; w_c \hat{\mathbf{h}}]}{\|[w_g \hat{\mathbf{p}}; w_c \hat{\mathbf{h}}]\| + \epsilon} \in \mathbb{R}^{384}, \quad w_g=0.4, \quad w_c=0.6. \quad (6)$$

Colour is up-weighted ($w_c > w_g$) because it is more discriminative for re-identification after background-heavy occlusion.

3.2 RAM Admission and DRM Promotion

RAM admits a box b only when it is geometrically coherent with the previous estimate $\hat{b}_{t-1}$:

$$\text{IoU}(b, \hat{b}_{t-1}) \geq \tau_{\text{ram}} \text{ and } \frac{|a(b) - \bar{a}_t|}{\bar{a}_t + \epsilon} \leq \tau_a, \quad (7)$$

with $\tau_{\text{ram}}=0.40$ and $\tau_a=0.25$. The area gate prevents distractor-driven size jumps from corrupting the RAM model.

DRM promotion requires consensus across the last $W=5$ RAM entries:

$$\sum_{j=t-W+1}^t \mathbf{1}[\cos(\phi_j, \phi_t) \geq \tau_{\text{sim}}] \geq m_{\text{min}}, \quad (8)$$

with $\tau_{\text{sim}}=0.60$ and $m_{\text{min}}=3$, filtering entries captured during brief blur or partial occlusion. Each

DRM anchor stores descriptor ψ_k and promotion timestamp ρ_k .

4 Occlusion Handling and Recovery

4.1 Occlusion Entry

The tracker declares occlusion when $s_t < \tau_{occ}=0.55$ persists for $N_p=1$ consecutive frame(s), at which point the last bounding-box estimate is frozen and the recovery pipeline takes over.

4.2 Ego-Motion Compensation

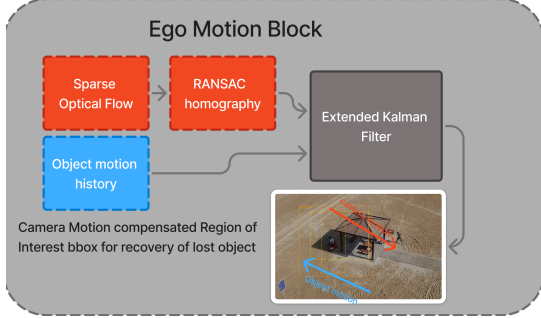


Figure 4: Ego-Motion Block. Sparse LK optical flow on background keypoints feeds RANSAC homography estimation; the resulting transform warps the EKF centre prior to kinematic integration, yielding a camera-motion-robust recovery search window.

As shown in Fig. 4, SiamRAM estimates camera motion every frame and feeds it into the EKF to keep the recovery search window anchored near the target’s true re-entry location.

Homography estimation. Sparse Lucas–Kanade flow [1] on Shi–Tomasi background keypoints yields a 3×3 projective transform H_t via RANSAC, flagged reliable when the inlier ratio exceeds 0.50.

EKF. A 4D state $[c_x, c_y, v_x, v_y]^T$ tracks the target centre and velocity. At each prediction step the centre is first warped by H_t (when reliable) to remove camera-induced displacement, then advanced by the kinematic velocity (v_x, v_y) . Box size is smoothed separately with $\alpha_s=0.85$. During occlusion the EKF runs prediction-only, providing the centre anchor for the search ROI.

4.3 Multi-Phase Reacquisition Pipeline

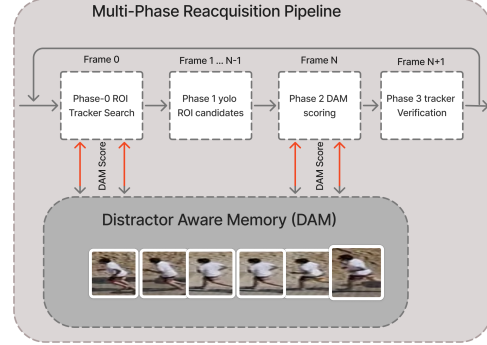


Figure 5: Multi-Phase Reacquisition Pipeline. Phase 0 runs SiamABC within the EKF-predicted ROI; Phase 1 collects YOLO candidates across multiple frames to enable velocity estimation; Phase 2 scores candidates via DAM appearance similarity and velocity consistency; Phase 3 verifies the top candidate with the Siamese tracker before resuming normal tracking.

Once occlusion is declared, SiamRAM runs a four-phase cascade illustrated in Fig. 5. Each phase is attempted in turn; the first to succeed reinitialises the tracker and resumes normal tracking.

Phase 0 — Siamese ROI Search. SiamABC is seeded at the EKF-predicted centre with the target’s median size and run within the EKF ROI. If the score $s_t \geq 0.80$ and a lightweight DRM check (augmented with a single-frame direction score) passes, recovery succeeds immediately—without invoking YOLO.

Phase 1 — YOLO Candidate Collection. YOLOv11n is run on an expanded crop centred on the EKF box across $F_c=3$ consecutive frames, building per-candidate velocity tracklets. The search margin grows as:

$$\delta_t = \min(\delta_1 \cdot g^{\lfloor n/T_g \rfloor}, \delta_{\max}), \quad (9)$$

with $\delta_1=100$ px, $g=1.4$, $T_g=150$, $\delta_{\max}=500$ px for standard objects, and tighter parameters ($\delta_1=20$, $g=1.3$, $T_g=50$, $\delta_{\max}=40$ px) for tiny objects ($< 3\%$ of frame area) to avoid false positives from background clutter.

Phase 2 — DAM Appearance and Velocity Scoring. Each multi-frame-tracked candidate c_i is scored against DRM anchors:

$$\mathcal{S}^*(c_i) = \max_k \cos(\psi_k, \phi(I_t, c_i)). \quad (10)$$

To reject appearance-matching distractors that move inconsistently with the target, a velocity consistency term is computed from the candidate’s camera-compensated velocity $\mathbf{v}_{c_i}$ versus the EKF velocity $\mathbf{v}_{\text{ekf}}$:

$$\mathcal{V}(c_i) = 0.8 \cdot \frac{\cos(\mathbf{v}_{\text{ekf}}, \mathbf{v}_{c_i}) + 1}{2} + 0.2 \cdot s_{\text{spd}}, \quad (11)$$

$$s_{\text{spd}} = \text{clip}\left(1 - \left| \frac{\|\mathbf{v}_{c_i}\|}{\|\mathbf{v}_{\text{ekf}}\|} - 1 \right|, 0, 1\right), \quad (12)$$

with $s_{\text{spd}}=0$ if $\cos < 0$, and a hard gate clamping $\mathcal{V} \leq 0.05$ when $\cos < -0.5$. The composite ranking score is:

$$\tilde{\mathcal{S}}(c_i) = \mathcal{S}^*(c_i) + \lambda_{\text{vel}} \mathcal{V}(c_i), \quad \lambda_{\text{vel}}=0.20. \quad (13)$$

When $\|\mathbf{v}_{\text{ekf}}\|$ falls below a minimum threshold, $\mathcal{V}$ is set to a neutral 0.5.

Phase 3 — Tracker Verification. The top DAM-accepted candidate is position-shifted by one EKF-velocity step (to account for the collection lag), used to reinitialise SiamABC, and verified with one forward pass. A score ≥ 0.70 confirms reacquisition; failure returns to Phase 0.

5 Conclusion

SiamRAM combines online template adaptation, test-time batch normalisation, a dual-buffer appearance memory with geometrically gated admission and consensus promotion, ego-motion-compensated Kalman filtering, and a four-phase reacquisition cascade. The velocity consistency term adds a motion-domain discriminator that penalises appearance-matched distractors whose kinematics conflict with the EKF prediction, providing a complementary signal beyond appearance alone.

References

- [1] B. D. Lucas and T. Kanade, “An iterative image registration technique with an application to stereo vision,” *IJCAI*, pp. 674–679, 1981.
- [2] L. Bertinetto et al., “Fully-convolutional siamese networks for object tracking,” *ECCV Workshops*, 2016.
- [3] PinataFarms, “FEAR: Fast, Efficient, Accurate, and Robust Visual Tracker,” *ECCV*, 2022.
- [4] *Author(s) TBD*, “SiamABC: Siamese Tracking with Adaptive Batch Calibration,” *arXiv*, 2025.
- [5] S. M. Raza et al., “EdgeDAM: Real-time Object Tracking for Mobile Devices,” *arXiv:2603.05463*, 2026.
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- [7] R. E. Kalman, “A new approach to linear filtering and prediction problems,” *J. Basic Engineering*, vol. 82, pp. 35–45, 1960.